

UBA Keystore Validation Script - Usage Guide

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UBA Keystore Validation Script - Usage Guide

Overview

The **keystore_validation.sh** script is a comprehensive tool for validating UBA (User Behavior Analytics) keystores and certificates. It performs thorough validation of all certificate-related components in a UBA deployment, including keystores, truststores, certificate files, and trust relationships.

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Prerequisites

- **Root Access:** The script must be run as root to access UBA configuration files and use **sudo -u caspida**
- **Java Environment:** java must be installed with keytool available
- **UBA Installation:** UBA must be installed in the expected location (**/opt/caspida**)

Command Line Options

```
Usage: ./keystore_validation.sh [OPTIONS]
```

Options:

- | | |
|----------------------|--|
| -j, --java-home PATH | Set JAVA_HOME path (default: auto-detect) |
| -s, --summary | Generate machine-readable summary report |
| --skip-jvm-full | Skip full JVM truststore validation (faster) |
| -h, --help | Show this help message |
| --keytool-help | Show detailed keytool command examples |

Option Details

- **-j, --java-home PATH**: Specify a custom JAVA_HOME path. The script will auto-detect if not provided.
- **-s, --summary**: Generate a machine-readable JSON summary report at the end of validation.
- **--skip-jvm-full**: Skip comprehensive JVM truststore validation for faster execution.
- **-h, --help**: Display usage information and exit.
- **--keytool-help**: Display detailed keytool command examples and exit.

Usage Examples

Basic Validation

```
sudo ./keystore_validation.sh
```

Performs complete validation with auto-detected Java environment.

Custom Java Home

```
sudo ./keystore_validation.sh --java-home /usr/lib/jvm/java-11-openjdk
sudo ./keystore_validation.sh -j /opt/java
```

Specify a custom Java installation path.

Generate Summary Report

```
sudo ./keystore_validation.sh --summary > keystore_report.json
```

Generate a machine-readable summary report and save to file.

Fast Validation

```
sudo ./keystore_validation.sh --skip-jvm-full
```

Skip full JVM truststore validation for faster execution.

Help and Documentation

```
./keystore_validation.sh --help
./keystore_validation.sh --keytool-help
```

Display usage help or detailed keytool command reference.

Output Examples

Basic Validation Output

When running basic validation, the script provides comprehensive output:

```
[2025-06-18 15:08:31] UBA Keystore and Certificate Validation
[2025-06-18 15:08:31] =====
JAVA_HOME: /usr/lib/jvm/java-1.8.0-openjdk
CASPIDA_HOME: /opt/caspida

[2025-06-18 15:08:31] Validating Critical UBA UI Authentication Certificates
=====
  Checking UBA UI authentication certificate configuration...

  ✓ Found: ui.auth.rootca=/var/vcap/store/caspida/certs/my_certs/my-root-ca.crt.pem
[SUCCESS]    ✓ Certificate file exists: /var/vcap/store/caspida/certs/my_certs/my-root-ca.crt.pem
[SUCCESS]    ✓ Valid certificate: expires Jun 12 00:23:28 2026 GMT
  □ Cross-validating against UBA keystores...
    o Certificate not in keystore: uba-keystore
    o Certificate not in keystore: keystore.jm
    o Certificate not in keystore: server.keystore.jks
[Δ] VALIDATION: UI certificate NOT found in any UBA keystore
  This may be normal if UI uses separate certificate files
  Recommendation: Verify UBA web service is using the configured certificate

[2025-06-18 15:08:39] Validating UBA Unified Keystore
  Path: /opt/caspida/conf/keystore/uba-keystore
  User: caspida
[SUCCESS] UBA Unified Keystore validated successfully (4 certificates)
  Certificate: jmserver
  Certificate: uba-server
  Certificate: kafka-server
  Certificate: root-ca
```

Key Features Validated

1. **UBA UI Authentication Certificates**
 - Root CA certificate validation
 - Server certificate validation
 - Private key validation
 - Cross-validation against UBA keystores
2. **UBA Unified Keystore**
 - Accessibility and password validation
 - Certificate count and details
 - Expiration date checking
3. **JobManager Keystore**
 - Keystore integrity
 - Certificate validation
4. **Kafka SSL Certificates**
 - Server keystore validation
 - Truststore validation
 - Certificate chain verification
5. **Java System Truststore**
 - JVM cacerts validation
 - Trust relationship verification

Success Indicators

- ✓ Green [SUCCESS] messages indicate passed validations
- Certificate counts show keystore contents
- Expiration dates are checked automatically

Warning Indicators

- ⚠ Yellow [⚠] messages indicate potential issues
- Recommendations provided for each warning
- Issues may be normal depending on configuration

Error Indicators

- ✗ Red [✗] messages indicate validation failures
- Detailed error descriptions provided
- Exit code reflects number of errors found

Script Features

Certificate Validation Capabilities

1. **Keystore Access Validation**
 - Tests multiple common passwords
 - Validates file permissions
 - Checks user access (caspid user)
2. **Certificate Content Validation**
 - Expiration date checking
 - Certificate chain verification
 - Fingerprint validation
3. **Trust Relationship Validation**
 - Cross-keystore certificate verification
 - Java truststore integration
 - SSL connection trust chains
4. **Configuration Validation**
 - UBA configuration file parsing
 - Certificate path resolution
 - Missing configuration detection

Performance Considerations

- **Full Validation:** Complete verification including JVM truststore (~2-5 minutes)
- **Fast Validation:** Skip JVM validation for faster execution (~30 seconds)
- **Summary Mode:** Generate machine-readable output for automation

Exit Codes

- **0**: All validations passed successfully
- **>0**: Number of validation errors encountered

Files and Paths Validated

UBA Keystores

- `/opt/caspida/conf/keystore/uba-keystore` - Unified UBA keystore
- `/opt/caspida/conf/jobconf/keystore.jm` - JobManager keystore
- `/opt/caspida/conf/kafka/auth/server.keystore.jks` - Kafka server keystore
- `/opt/caspida/conf/kafka/auth/server.truststore.jks` - Kafka truststore

Configuration Files

- `/opt/caspida/conf/deploy/conf-cache.properties` - Main configuration
- `/opt/caspida/conf/ui/application.properties` - UI configuration
- Various other UBA configuration files

Certificate Files

- UBA UI authentication certificates (configurable paths)
- System CA certificates
- Custom certificate installations

Security Considerations

1. **Privileged Access**: Script requires root access for complete validation
2. **Password Security**: Uses secure methods to test keystore passwords
3. **File Permissions**: Validates appropriate file access permissions
4. **Temporary Files**: Cleans up temporary files securely

Integration

The script can be integrated into: - **Deployment Pipelines**: Automated validation after UBA installation - **Monitoring Systems**: Regular certificate health checks - **Troubleshooting**: Diagnostic tool for certificate issues - **Compliance Audits**: Certificate inventory and validation

Related Documentation

- UBA Certificate Management Guide
- UBA Keystore Management
- Troubleshooting Guide

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